

WETCHARTS_downscaling.R

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Disaggregate_Wetcharts	<i>Create model-averaged downscaled Wetcharts wetland methane maps</i>
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Description

‘Disaggregate_Wetcharts’ writes 1 monthly netcdf file of gridded wetland methane emissions per model subset set in the config and per land dataset used for the downscaling. Includes optional visuals as well.

Usage

```
Disaggregate_Wetcharts(  
  input_directory,  
  output_directory,  
  domain,  
  verbose,  
  plot_directory,  
  inventory_year,  
  County_Tigerlines,  
  State_Tigerlines,  
  focus_city_tigerlines,  
  Use_NLCD,  
  Use_NALCMS,  
  NLCD_file,  
  NALCMS_file,  
  Wetcharts_model_subset  
)
```

Arguments

input_directory	Character providing the full filepath to save/load input data
output_directory	Character providing the full filepath to save processed data
domain	SpatRaster providing the desired output grid, including the desired resolution and coordinate reference system
verbose	Logical indicating whether to save additional output. This includes monthly plots of the gridded methane emissions on log scales, saved separately for each model subset and each land cover dataset used.
plot_directory	Character providing the full filepath to save figures. Only relevant if verbose = TRUE.
inventory_year	Character indicating the desired year of data to use.
County_Tigerlines	SpatVector. United States Census Bureau county shapefile. Available at https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html . Only relevant if verbose=TRUE.
State_Tigerlines	SpatVector. United States Census Bureau county shapefile. Available at https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html . Only relevant if verbose=TRUE.
focus_city_tigerlines	SpatVector. United States Census Bureau county shapefile. Available at https://www.census.gov/geographies/mapping-files/time-series/geo/tiger-line-file.html . Only relevant if a focus city was set in main and verbose=TRUE.
Use_NLCD	Logical. Pulled from config file. Indicating whether to use the National Land Cover Database (NLCD) to downscale Wetcharts.
Use_NALCMS	Logical. Pulled from config file. Indicating whether to use the North America Land Change Monitoring System (NALCMS) to downscale Wetcharts.
NALCMS_file	Character. The NALCMS land cover data as a tif. Available at http://www.cec.org/north-american-land-change-monitoring-system/ .
Wetcharts_model_subset	Numeric list. Pulled from config file. Indicates which models of Wetcharts to average across. Multiple list entries are allowed, providing multiple variations of wetcharts.

Value

Nothing is returned from the function, but the main outputs are monthly netcdf files of the methane emissions from wetlands with 1 file per land cover used and per model subset. They are titled "Wetcharts_landcover_Downscaled_subset_A.nc" where landcover is NLCD or NALCMS and A is just a numeric to identify which model subset, in case multiple were set.

If verbose is set to TRUE, then multiple figures are also saved. Log scale plots with consistent axes are saved for each month, model subset, and land cover used. They are named "Wetcharts_landcover_subset_A_month_B" where landcover is NLCD or NALCMS, A is a numeric to identify the model subset, and B is the month (1-12).

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Examples

```
library(terra)
grid_bbox=cbind(c(-76.65,-73.65),c(38.97,40.97))
grid_res=0.01
grid_crs="epsg:4326"
grid <- rast(nrows=diff(range(grid_bbox[,2]))/grid_res,
             ncols=diff(range(grid_bbox[,1]))/grid_res, xmin=min(grid_bbox[,1]),
             xmax=max(grid_bbox[,1]), ymin=min(grid_bbox[,2]), ymax=max(grid_bbox[,2]),
             crs=grid_crs)

Urban_Tigerlines <- vect("~/../Desktop/Urban_Tigerlines/tl_2018_us_uac10.shp")
focus_city <- terra::subset(Urban_Tigerlines,Urban_Tigerlines$NAME10 %in% "Philadelphia, PA--NJ--DE--MD")
Disaggregate_Wetcharts(input_directory=~../Desktop/input/",
                       output_directory=~../Desktop/",
                       domain=grid,
                       verbose=TRUE,
                       inventory_year=2018,
                       plot_directory=~../Desktop/plots/",
                       State_Tigerlines=vect("~/../Desktop/State_Tigerlines/tl_2018_us_state.shp"),
                       County_Tigerlines=vect("~/../Desktop/County_Tigerlines/tl_2018_us_county.shp"),
                       focus_city_tigerlines=focus_city,
                       Use_NLCD=TRUE,
                       Use_NALCMS=TRUE,
                       NLCD_file=file.path("~/../Desktop/nlcd_2019_land_cover_148_20210604/nlcd_2019_land_cover_148_20210604.shp"),
                       NALCMS_file=file.path("~/../Desktop/NALCMS_2020_land_cover/NA_NALCMS_landcover_2020_30m/data/NA_NALCMS_landcover_2020_30m.shp"),
                       Wetcharts_model_subset=list(c(1913,1914,1923,1924,1933,1934,2913,2914,2923,
                                                       2924,2933,2934,3913,3914,3923,3924,3933,3934)))
```

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